# AI Grader — New Result Experience

## 1. Overview

Today, when a student's subjective test is evaluated, they see a plain score and generic text feedback (e.g. "Introduction was largely relevant, avoid abbreviations") — with no visual link between the feedback and the actual part of their answer it refers to.

We are building a new **annotation-based Result Experience**: teacher-style tick/cross marks and red/green/yellow highlight boxes placed directly on the student's answer. Tapping or scrolling to a highlight opens a feedback box explaining that specific mark. Students can also switch to a **Model Answer** tab to compare.

Two tracks run in parallel:

- **DS** decides *what* the annotation is — tick, cross, how many marks, which part of the answer it applies to.
- **Tech** builds *how* it's shown — rendering highlights at the right position, opening feedback on interaction, the Model Answer tab, and consuming whatever DS sends.

## 2. Problem

- Students have to zoom through the answer PDF to read it, especially on mobile.
- Feedback is generic and not mapped to the specific part of the answer it refers to.
- Students have to interpret on their own what was right, wrong, or needs improvement.
- No teacher-style visual cues (highlights, ticks, colour-coding) exist today.

## 3. Scope — Phase 1

| In Scope |
| --- |
| All Tests → Pure AI → Single-question subjective tests |
| AI Grader Widget → Multi-question AI-evaluated tests |
| Web and App platforms |

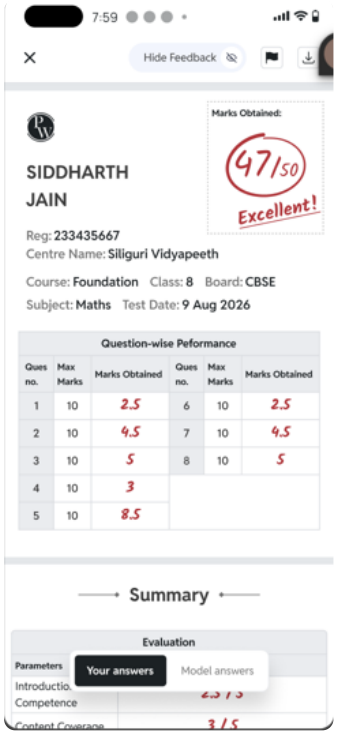
Phase 1 covers Pure AI tests (online, in-app submission). Subjective and Hybrid tests, the offline flow and SEP evaluation changes are covered in Phase 2 PRD (see Section 7). Everything other than the Result Experience continues to work exactly as it does today.

## 4. Design Reference

Figma: AI Grader Main — Result Experience

Key elements from the design:

- **Cover page** — Student details + Scores + question-wise marks table.



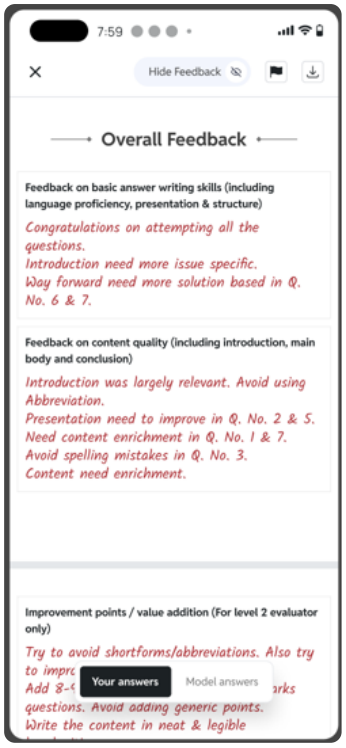

Cover page

- **Parameter-wise Marks table** — applicable in UPSC only.




Parameter Level Feedback

- **Overall summary page** — feedback given on test level, applicable only in UPSC.



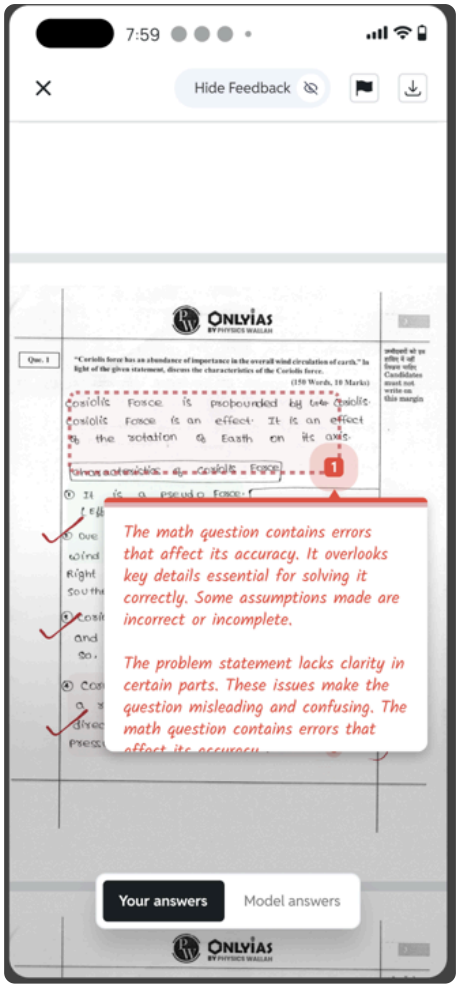

Overall Feedback

- **Colour-coded highlight boxes** — numbered boxes on the answer, each paired with a matching feedback comment. Colour meaning is fixed (see 4.1 below).

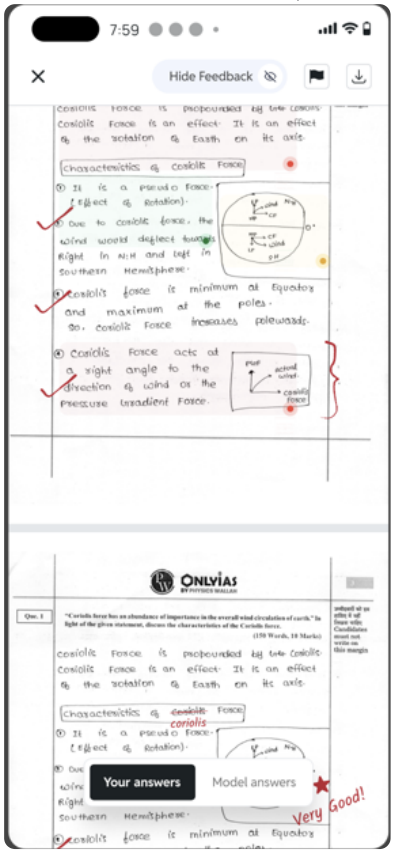
#### 4.1 Highlight Colour Significance

| Colour | Meaning |
|---|---|
| **Red** | Mistake — this part of the answer is incorrect. |
| **Yellow** | Needs improvement — not wrong, but could be better. |
| **Green** | Correct — this part of the answer is right. |

- **Tick marks / circled scores** — used for simple correct/incorrect answers that don't need a detailed comment.




Tick mark example




Tick mark — spelling mistake case

- Toggle: **Your answers ↔ Model answers**
- Toggle: **Hide Feedback ↔ Show Feedback**

- Download — 3 separate options: **My Answer Paper**, **Evaluated Answer Paper**, **Model Answer Paper**
- Report — on the **complete evaluation**, or on an **individual evaluation** (a specific question/highlight)

#### 4.2 Screen Flow by Cohort

The result screen sequence differs by cohort:

| UPSC | Foundation |
|---|---|
| 1. Cover page | 1. Cover page |
| 2. Parameter-wise marking | 2. Student answer copy with annotations |
| 3. Overall summary page | |
| 4. Student answer copy with annotations | |

#### 4.3 Cover Page — Data Points & Rules

**Goal:** give the student a quick summary of the result: total marks, remark, and marks in each question.

**A. Student details and score:**

**Figma Link-** https://www.figma.com/design/6HP4K6drY8YNjxNYT8fRdC/Untitled?node-id=23-2515&t=qnvshoCIqcUsm3gF-0 Connect your Figma account

| Data point | Foundation | UPSC | Example |
|---|---|---|---|
| Student name | ✅ | ✅ | SIDDHARTH JAIN |
| Marks obtained | ✅ | ✅ | 47 (decimals allowed, e.g. 47.5) |
| Total marks | ✅ | ✅ | 50 |
| Remark | ✅ | ✅ | Excellent! |
| Reg. no. | ❌ | ❌ | 233435667- data not present |
| Course | ✅ | ❌ | Foundation |

| | | | |
|---|---|---|---|
| Class | ✅ | ❌ | 8 |
| Board | ✅ | ❌ | CBSE |
| Subject | ❌ | ❌ | Maths- one test can have multiple subejcts |
| Test name | ❌ | ✅ | Simulator Test 04 |
| Contact no. | ❌ | ✅ | 9090909090 |
| Test date | ✅ | ✅ | 9 Aug 2026 |

**B. Remark logic**

- The remark is derived from `marks obtained / total marks` (%). It is not hardcoded.
- Good marks → positive remark. Low marks → encouraging remark.
- Each cohort can choose whether to show the remark or not (configurable per cohort).

| Marks % | Remark | Tone |
|---|---|---|
| 90% and above | Excellent! | Positive |
| 60% to 89% | Very Good! | Positive |
| 40% to 59% | Good Effort! | Encouraging |
| Below 40% | Keep Practicing! | Encouraging |

**C. Question-wise performance table**

- Columns: Ques no., Max marks, Marks obtained.
- Number of rows = number of questions in the test. The table extends automatically as the question count grows (continues on the next page if needed).
- A single-question test still shows the table, with 1 row.
- An unattempted question shows `0`.
- Marks format: Marks obtained are shown in red, handwritten-style font.

**D. Data consistency**

- Marks obtained at the top = sum of marks obtained across all questions.
- Total marks at the top = sum of max marks across all questions.

**E. Fixed elements**

- PW logo

**F. Not on the cover page**

Parameter-wise marks and Overall feedback are separate pages, UPSC only (see 4.2).

#### 4.4 Types of Annotations

The student's answer copy carries two types of annotations.

| Type | What it looks like | Feedback box? | Use it when |
|---|---|---|---|
| 1. **Highlighted bounding box** | A numbered, coloured box (red / yellow / green) drawn around a part of the answer | Yes. Every box has a feedback comment. | A specific part of the answer needs an explanation |
| 2. **Handwritten teacher-style** | Tick, cross, underline, circled score and stamps, drawn like a teacher marking with a pen | No. It is a quick visual cue. Only underline can optionally carry a short comment. | A simple correct / incorrect / quick remark is enough, with no detailed comment |

##### Type 1: Highlighted bounding box

- A box is placed on the exact part of the answer it evaluates.
- Colour can be **red, yellow or green**. Colour meaning is fixed (see 4.1).
- Each box has a **number** and a **feedback comment** with the same number and colour. Tapping and scrolling the box opens its feedback (see User Story 1).
- **Every box must have feedback.** A box without feedback is not allowed. If no comment is needed, use a Type 2 mark instead.
- **Phase 1:** view only for the student.
- **Phase 2:** editable by the evaluator on SEP. The evaluator can add, move, resize or remove a box, change its colour, and edit its feedback (see 7.1).

**Highlighted Bounding box:**

The meaning is the **same for all cohorts** (UPSC and Foundation). Only the tone of the language can differ by cohort: Foundation feedback is simple and encouraging, UPSC feedback is analytical and exam-oriented.

| Box colour | Polarity | Meaning | Example feedback |
|---|---|---|---|
| **Green** | Positive | Confirms that this part of the answer is correct or well done | "Correct definition of Coriolis force, with the right cause." |
| **Red** | Negative | Points out an error, a wrong fact or a required point that is missing | "Coriolis force is minimum at the equator, not maximum. Check this point." |
| **Yellow** | Neutral | A suggestion or a note. Not wrong, but could be better. No right / wrong verdict. | "Point is right. Add an example of a cyclone to make it stronger." |

**Rules for feedback polarity**

- If one comment has both praise and a mistake, DS splits it into two boxes, one for each point.
- Polarity is read from the box colour when we study report data. For example, a thumbs down on a red box means the student disagrees with the criticism.
- **Phase 2:** if the evaluator changes the colour of a box on SEP, the polarity changes with it.

#### Type 2: Handwritten teacher-style marks

These look like a teacher marking on paper. They have no numbered box.

| Mark | Use case | Example |
|---|---|---|
| **Tick (✓)** | The point, step or answer is correct | Correct definition of Coriolis force, MCQ answer matches the key |
| **Cross (✗)** | The point, step or answer is wrong | Wrong formula or wrong final value |

| | | |
|---|---|---|
| **Circled score** | Marks given for that question or part of the answer, written in a circle next to it | 2.5 circled at the end of Q1 |
| **Underline** | Marks a specific word or line. **It can have a comment or not.** Red for a mistake, green for a good point worth noticing. | Wrong word underlined in red with a short comment, key term underlined in green with no comment |
| **Strike-through** | Content that is wrong or not needed | Irrelevant line that adds no value |
| **Text stamps** | Short remark stamped on the answer or at the end of a question. Proposed set: *Good*, *Very Good*, *Incomplete*, *Irrelevant*, *Needs more detail*. | "Very Good" stamped at the end of a strong answer |

**Rules for Type 2 marks**

- They do not open a feedback box. They only show the mark in place.
- **Exception: Underline** can optionally carry a comment. If a comment is added, it is shown with the underline. If not, the underline is shown alone.
- They are positioned on the exact part of the answer they refer to, and stay in place while scrolling and across multi-page answers.
- If a part of the answer needs a detailed explanation, DS uses a Type 1 box instead of a Type 2 mark.
- Both types can appear on the same answer.
- **Phase 2:** Type 2 marks are also editable on SEP, same as Type 1. The evaluator can change, update or remove an existing mark and add more marks.

**Text stamps: criteria and when a stamp is given**

- A stamp can be given once per question, only wherever required. It is not given for each part or each point.
- An unattempted question gets no stamp (its marks show `0` on the cover page).
- Objective-type answers that only need a tick or cross do not get a stamp.

| Check order | Stamp | Given when | Example |
|---|---|---|---|
| 1 | **Irrelevant** | The answer or a point does not relate to the question asked. None of the key points of the model answer are present. | Question is on the Coriolis force, but the student wrote about tidal energy |
| 2 | **Incomplete** | The answer or point is relevant, but one or more required parts of the question are missing, or the answer stops in the middle, or something major is missing in the content. | A 3-part question where only 2 parts are answered |
| 3 | **Very Good** | The student has written a very strong point, or has performed very well at question level. | 8.5 out of 10 |
| 4 | **Good** | The answer is relevant and complete, and the student scored 60% to 79% of the question's marks. | 6.5 out of 10 |

**Phase 2:** the evaluator can change or remove the stamp on SEP.

#### 4.5 Result Screen Controls

These controls are on the result screen. Look and behaviour follow the Figma design.

| Control | What it does |
| --- | --- |
| **Tabs** | The screen has 2 tabs:<br>1. **Answer copy:** the student's own answer copy with all annotations (Type 1 and Type 2).<br>2. **Model Answer:** the model answers for all questions of the test, shown together in one complete view. |
| **Hide / Show Feedback toggle** | **Hide Feedback:** hides all annotations on the answer copy, both Type 1 (boxes) and Type 2 (handwritten marks). The student sees the clean answer.<br>**Show Feedback:** brings back all annotations, both types. |
| **Report button** | Sits next to the Hide / Show Feedback toggle. The student can report an issue on the complete evaluation. The existing report design is reused. |
| **Download button** | The student can download 3 separate files, as per the Figma design:<br>• **My Answer Paper:** the student's answer copy without any annotations.<br>• **Evaluated Answer Paper:** the answer copy with all annotations, as designed in Figma.<br>• **Model Answer Paper:** the model answers that we create for the test. |
| | |

| | |
|---|---|
| **Report on a highlighted box comment** | The student can also report on the comment inside a highlighted box (Type 1). The report is captured against that specific evaluation, not the whole test. |

#### 4.6 Report button

**Goal:** collect a simple signal from students on whether the evaluation is right and useful. Every option is written so that it points to one clear thing that Product, DS or Tech can act on.

| | A. Feedback-level report |
|---|---|
| **Where** | Inside each feedback box (Type 1 highlight) |
| **Question it answers** | Is this specific feedback right and useful? |
| **Main user of the data** | DS (which part of the AI evaluation to fix) |
| **Format** | Thumbs up / thumbs down. A short box opens with options + optional comment + Submit. |

Inside each feedback box (Type 1 highlight), the student sees thumbs up and thumbs down.

- **Thumbs up:** saved on tap. No reasons are asked.
- **Thumbs down:** saved on tap. A box opens with **"What can be improved?"**, the reasons below (multi-select), an optional comment and Submit.

| # | Thumbs down reason | Covers |
|---|---|---|
| 1 | Feedback is wrong | Feedback quality (accuracy) |
| 2 | Feedback is too generic | Feedback quality (specificity) |
| 3 | Hard to understand feedback | Feedback quality (clarity) |
| 4 | Feedback is on the wrong point | Mapping |

| | | |
|---|---|---|
| 5 | Highlight colour is wrong | Polarity |
| 6 | Highlight is misplaced | Mapping |
| 7 | Other | Comment is mandatory |

**Rules**

- Submit is enabled after at least one reason is selected.
- One reaction per student per feedback. The student can change the thumb later.
- Thumbs are only on Type 1 boxes.
- The bottom sheet opens on the same screen. The student does not lose their place in the answer.

##### Data captured for every thumbs reaction

Tech saves these fields for each reaction. The student only taps the thumb, picks options and types a comment. Everything else is attached automatically.

| Field | Value | Source |
|---|---|---|
| Student name | Student name | Automatic |
| Question ID | The question the feedback belongs to | Automatic |
| Feedback ID | The ID of the feedback comment (same as the annotation ID sent by DS) | Automatic |
| Reaction | Thumbs up or thumbs down | Student |
| Options chosen | The options the student selected | Student |
| Comment | Free text, if the student wrote anything | Student |
| Test ID, cohort, platform, time of reaction | Test ID, UPSC / Foundation, Web / Android / iOS, timestamp | Automatic |
| | | |

| | | |
|---|---|---|
| Feedback details | Box colour (green / yellow / red, which gives the polarity), annotation type, page number, feedback text shown | Automatic (from DS output) |

#### 4.7 Rating

A star rating popup is shown when the student exits the result experience (back, close or any CTA) from any screen.

- The student picks 1 to 5 stars. Submit is disabled until a star is given.
- After the star, a heading and options appear (multi-select, optional). If *Other* is selected, the comment is mandatory.
- Once the student has rated, the popup is not shown again until they **reattempt the test**.
- If the student closes the popup without Submit, nothing is saved and they continue to where they were going.

| Stars | Heading | Options |
|---|---|---|
| **1–2 stars** | Your feedback will help us improve. What should we fix first? | • Marks are not correct<br>• Feedback not helpful<br>• Model answer incorrect<br>• Uploading file issue<br>• Poor overall experience<br>• Other |

| | | |
|---|---|---|
| **3 stars** | You're helping us get better. What can we improve? | • Marking more accurate<br>• Feedback more detailed<br>• Improve model answers<br>• Better overall experience<br>• Faster evaluation<br>• Other |
| **4–5 stars** | Glad it helped! What worked best for you? | • Accurate marking<br>• Quality of feedback<br>• Helpful model answers<br>• Great overall experience<br>• Fast evaluation<br>• Other |

### 5. User Stories, Requirements & Acceptance Criteria

#### User Story 1: Understand my evaluation in context

As a student, I want feedback mapped to the specific highlighted part of my answer, so that I understand exactly why I got or lost marks without interpreting it myself.

**Functional Requirements**

Highlight Rendering

1. Each evaluated region shall be marked with a numbered, colour-coded highlight box, positioned against the specific part of the answer it evaluates.
2. Simple correct/incorrect answers with no detailed comment shall use a tick mark or circled score instead of a numbered highlight box.

Feedback Interaction

1. Each numbered highlight shall have a corresponding feedback comment, shown with the same number and colour as the highlight it belongs to.
2. Feedback shall be viewable in context at the point of the highlight (per Figma interaction), without navigating away from the answer view.

3. Where multiple highlights exist on one answer, each highlight-comment pair shall remain uniquely identifiable — no colour or number reused within the same answer.
4. Where multiple highlights exist on one answer, they shall open/expand in the order of their position on the answer sheet (the highlight positioned first, top-to-bottom, opens first) — not in an arbitrary or reverse order.

**Acceptance Criteria**

AC1: Highlight-Feedback Match Given an answer has a highlighted region with feedback, When the student views the result, Then the highlight's number and colour shall match its feedback comment exactly.

AC2: Multiple Highlights — Uniqueness & Order Given multiple highlighted regions exist on one answer, When the student views the result, Then each highlight-comment pair shall remain uniquely identifiable, and shall open in the order of their position on the answer — the highlight positioned first shall open first.

AC3: Tick/Circled Score Fallback Given an answer needs no detailed feedback, When evaluated, Then a tick mark or circled score shall display without a highlight box.

#### User Story 2: Review my evaluation easily on mobile

As a student, I want to scroll through my annotated answer sheet smoothly on my phone, so that I don't lose track of which comment belongs to which part of my answer.

**Functional Requirements**

Scroll & Pairing

1. Highlight-comment pairing shall remain correct as the student scrolls, including across multi-page answer sheets.
2. Highlight numbering shall not repeat or mismatch across pages within the same answer sheet.

Persistent Controls

1. The Download control and the Hide/Show Feedback toggle shall remain accessible (e.g. persistent header) while the student scrolls.

Mobile Layout

1. The annotated answer view shall be readable on mobile screen widths without requiring horizontal zoom to identify a highlight's number or colour.

**Acceptance Criteria**

AC1: Multi-Page Scroll Integrity Given a multi-page answer sheet, When the student scrolls, Then highlights shall continue to map correctly to their comments with no mismatched numbering across pages.

AC2: Mobile Legibility Given a student is on mobile, When viewing any annotated page, Then highlight numbers and colours shall be legible without pinch-zoom.

#### User Story 3: Know the model answer, download, and report on my result

As a student, I want to see model answers, download my answer copy, and report a problem with my evaluation, so that I can verify my result and flag anything that looks wrong.

**Functional Requirements**

Your Answer / Model Answer Tabs

1. The result screen shall have two tabs: **Your Answer** and **Model Answer**.
2. The **Your Answer** tab shall show the student's own evaluated answer sheet — marks, highlights, and feedback (as in User Story 1).
3. The **Model Answer** tab shall show the model answer for every question in the test — whether or not the student attempted that question — with no annotations.

Download

1. Students shall be able to download three separate documents, independently of one another:
   - **My Answer Paper** — the student's original, as-submitted answer sheet, with no annotations.
   - **Evaluated Answer Paper** — the student's answer sheet with annotations, marks, and feedback.
   - **Model Answer Paper** — the model answer sheet for the full test.

Report

1. Students shall be able to report an issue on the **complete evaluation** (the whole test's result).
2. Students shall also be able to report an issue on an **individual evaluation** — a specific question/highlight — from within that question/highlight's context.
3. Both report types shall let the student add a reason/comment describing the issue.

**Acceptance Criteria**

AC1: Tab Switch Given a student is on the result screen, When they switch between "Your Answer" and "Model Answer," Then the Your Answer tab shall show their own evaluated answer, and the Model Answer tab shall show every question's model answer with no annotations.

AC2: Three Independent Downloads Given a student opens Download, When they choose an option, Then they shall see and be able to independently download My Answer Paper, Evaluated Answer Paper, and Model Answer Paper.

AC3: Report — Complete Evaluation Given a student wants to flag an issue with their overall result, When they use Report at the test level, Then the report shall be captured against the complete evaluation.

AC4: Report — Individual Evaluation Given a student wants to flag an issue with one specific question/highlight, When they use Report from within that question/highlight, Then the report shall be captured against that specific evaluation, not the whole test.

## 6. User Flow — Answer Upload & Quality Check

#### 6.1 Flow

1. Student uploads their answer (image or PDF) for a question.
2. Student clicks **Submit** on that question.
3. **Existing check runs** (no change — already live for Pure AI Evaluated and Hybrid): the system checks that the answer is relevant, and that the image has no error / is not blurry.
4. **If an issue is found** → student is asked to re-upload (existing behaviour).
5. **If no issue is found** → the answer is accepted, student moves on.
6. Steps 1–5 repeat for every question in the test.
7. Once all questions are answered, student submits the **full test**.
8. AI evaluation runs on the submitted test.
9. **New Result Experience** is shown to the student.

*Same flow applies whether the test has a single question or multiple questions — each answer goes through the same check independently.*

#### 6.2 Functional Requirements

1. The existing relevancy/blur check on submit is unchanged for both Pure AI Evaluated and Hybrid flows — no new requirement here.

#### 6.3 Acceptance Criteria

AC1: Flow Ends in a Working New Result Experience Given a student completes upload → check → submit for every question and submits the full test, When AI evaluation finishes, Then the New Result Experience shall load, and the student shall be easily able to scroll to and click on a highlighted part to see and read its feedback (per User Story 1).

AC2: Works for Single or Multi-Question Tests Given a test has either a single question or multiple questions, When the student completes this flow, Then the New Result Experience shall render correctly in both cases.

## 7. Rollout

| Phase | Test type | Flow | Platforms |
| --- | --- | --- | --- |
| | | | |

| | | | |
|---|---|---|---|
| **Phase 1** | Pure AI — all tests | Online | All platforms |
| **Phase 2** | Subjective and Hybrid — all tests | Online and Offline | All platforms: Web, App (Android, iOS) |

#### 7.1 Phase 2 — Requirements

1. The New Result Experience shall be live for all **Subjective and Hybrid tests**, in both **online and offline** flows, single or multiple questions.
2. It shall work on all platforms: **Web and App (Android, iOS)**.
3. On **SEP**, the evaluator shall be able to **edit, remove, add and change** the evaluation (highlights, handwritten marks, marks, feedback) on any answer.
4. Every SEP change shall reflect in the student's New Result Experience.

#### 7.2 Phase 2 — Acceptance Criteria

AC1: Full Coverage Given any Subjective or Hybrid test (online or offline, single or multiple questions), When the student opens the result on Web, Android or iOS, Then the New Result Experience shall load.

AC2: SEP Changes Reflect Given an evaluator edits, removes, adds or changes an evaluation on SEP, When the student opens the result, Then the New Result Experience shall show the updated evaluation.

**End state: the New Result Experience is live on all subjective tests — online and offline, on Web, Android and iOS, including evaluation changes made on SEP.**